I. Introduction to Electronic Business

Definition:

Electronic Business (E-Business) means conducting ALL business activities - buying, selling, serving customers, collaborating with partners, managing internal operations - using the Internet and other electronic networks.

Meaning

- **E-Business** = use of information & communication technology (ICT) to run and improve the whole business.
- The term was popularised by IBM in 1997 ("e-business").
- It covers front-office (customers, sales) and back-office (production, finance, HR, supply chain).
- Business processes become digital, faster, cheaper and available 24 x 7.

Nature of E-Business

- **Digital:** work is done through computers, networks, web & mobile apps.
- **Borderless:** a small shop can reach customers in any country.
- **Information-driven:** data is the main resource; analytics guides decisions.
- **Automated:** orders, invoices, payments, stock updates happen with little manual work.
- **Interactive:** two-way communication with customers (chat, reviews, email).
- **Integrated:** links buyers, sellers, banks, logistics and government.
- **Customer-centric:** personalised offers, quick service, easy feedback.
- **Cost-effective:** no big shop rent, fewer middlemen, less paperwork.

★ Remember: E-business is the BROADER term - it includes e-commerce plus internal and partner processes.

2. Significance of E-Business

- **Global reach:** market grows from a city to the whole world.
- **Round-the-clock service:** the online store never closes.
- **Lower cost:** saves on rent, staff, printing, stock-holding.
- **Speed:** order → payment → delivery tracked in minutes.
- **Better customer service:** FAQs, chatbots, instant support, order tracking.
- **Competitive advantage:** small firms compete with big ones.
- **Data & decision making:** customer behaviour is recorded and analysed.
- **Employment & growth:** new jobs - web developers, digital marketers, delivery staff.
- **Supports Digital India:** UPI, ONDC, GeM, e-governance all rely on e-business ideas.

Functions of E-Business

- **Buying & selling** of goods, services and information online.
- **Marketing & advertising:** websites, SEO, social media, email.
- **Customer support:** help desk, chat, grievance handling.
- **Payment processing:** cards, net banking, UPI, wallets.
- **Order & inventory management:** stock levels updated automatically.
- **Supply chain & logistics:** supplier coordination, shipping, tracking.
- **Finance & accounting:** e-invoicing, GST filing, online banking.
- **HR & training:** online recruitment, payroll, e-learning.
- **Collaboration:** video meetings, cloud documents, intranet/extranet.

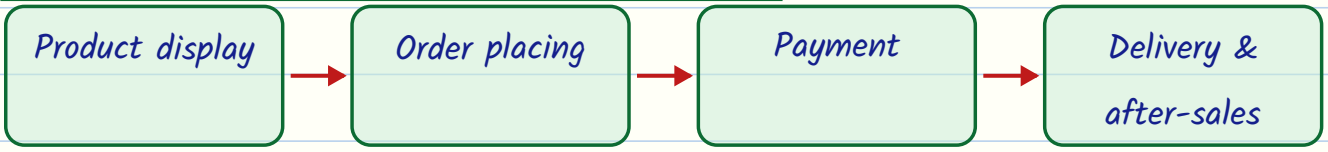
3. Definitions of Electronic Commerce

Electronic Commerce (E-Commerce):

The buying and selling of goods and services, and the transfer of money and data to complete these transactions, over the Internet or other electronic networks.

- Narrow view: only online selling & payment.
- Broad view: all commercial transactions based on electronic processing & transmission of data - text, sound, images.
- Shopping website, ticket booking, online banking, app-based taxi are all e-commerce.

Functions of Electronic Commerce (EC)



- **Information function:** catalogue, price, specifications, reviews.
- **Communication:** email, messaging, chat between buyer & seller.
- **Ordering:** shopping cart, order confirmation, invoice.
- **Payment function:** secure online payment gateways.
- **Delivery:** physical shipping, or digital download (e-books, software, tickets).
- **Customer service:** returns, refunds, warranty claims, tracking.
- **Market research:** collecting feedback, clicks, buying patterns.
- **Advertising & promotion:** banners, coupons, offers, loyalty points.

4. Advantages of E-Commerce

To Organisations (Sellers)

- Low set-up and running cost; no need for physical outlets.
- Access to national and global markets.
- Quick launch of new products and offers.
- Reduced paperwork and inventory cost (just-in-time).
- Customer data helps in personalised marketing.
- Improved brand image and 24 x 7 presence.

To Customers

- Shop anytime, anywhere - home, office, mobile.
- Wide choice and easy price comparison.
- Discounts, cashbacks, coupons.
- Home delivery saves time and travel.
- Reviews and ratings help to choose well.

To Society

- Fewer vehicles on roads → less pollution & traffic.
- Rural people get products and services (online education, telemedicine).
- Government services reach citizens easily.
- Generates employment and supports start-ups.

5. Disadvantages / Limitations of E-Commerce

Technical Limitations

- Needs internet connectivity and good bandwidth.
- Software / website crashes, slow servers, system compatibility problems.
- Security threats: hacking, phishing, data theft, card fraud.

Non-Technical Limitations

- Customer cannot touch, try or inspect the product.
- Delivery delays, wrong or damaged items, return hassles.
- Lack of trust - fake sellers, fake reviews.
- Privacy concerns - personal data is collected and may be misused.
- Digital divide - people without devices or skills are left out.
- Legal issues - taxation, jurisdiction, consumer protection across countries.
- Small shops may suffer from big online competitors.
- High competition and heavy advertising costs.

Quick Compare:

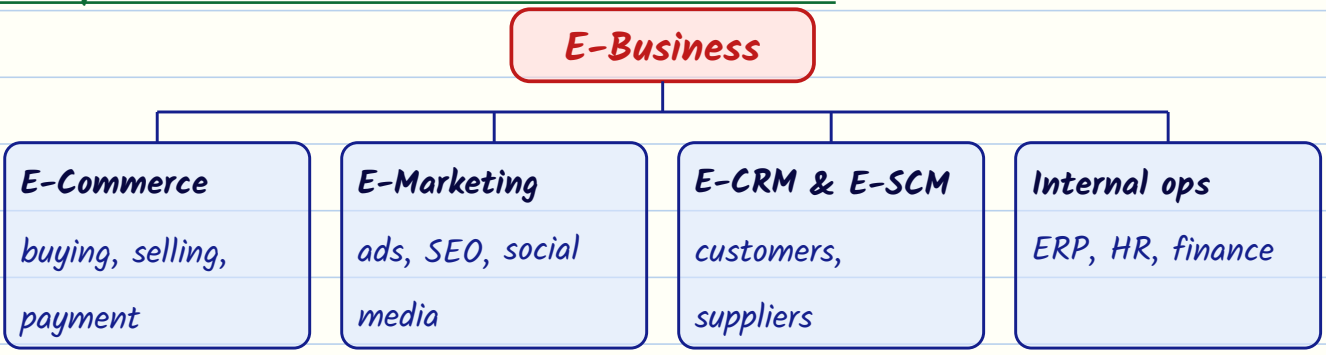
Advantage = convenience, low cost, global reach. Disadvantage = security risk, no physical check, dependence on technology.

6. E-Commerce vs E-Business

Basis	E-Commerce	E-Business
Meaning	Buying & selling online	All business activities done electronically
Scope	Narrow - a part of e-business	Broad - includes e-commerce
Focus	External transactions with customers	Internal + external processes
Parties	Buyer and seller	Customers, suppliers, employees, partners
Needs	Internet, website, payment gateway	Internet, intranet, extranet, ERP, CRM, SCM
Examples	Amazon shopping, IRCTC booking	ERP, online HR, supply chain, e-learning, plus selling
Money	Always involves a transaction	May or may not involve money

★ Key line: All e-commerce is e-business, but all e-business is not e-commerce.

Components of E-Business (the umbrella)



7. Internet Services

Internet:

A worldwide network of networks connected through TCP/IP that lets computers share information.

Main Internet Services used in E-Business

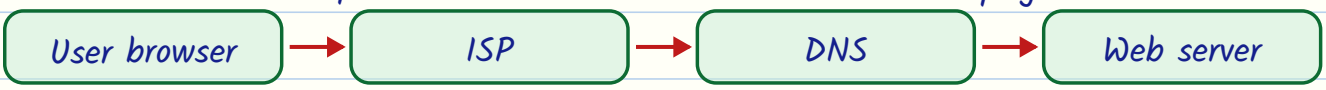
- **E-mail:** fast electronic messages and attachments; used for orders, invoices, notices.
- **World Wide Web (WWW):** web pages linked by hyperlinks, accessed by a browser (HTTP).
- **File Transfer (FTP):** upload / download large files between computers.
- **Telnet / SSH:** log in and control a remote computer.
- **Search engines:** Google, Bing - help customers find products & firms.
- **Chat / Instant messaging:** WhatsApp, live chat, chatbots for support.
- **Video conferencing:** Zoom, Meet - online meetings and demos.
- **VoIP:** voice calls over internet, cheap business calls.
- **Newsgroups / forums / blogs:** discussion, reviews and brand communities.
- **E-banking & payments:** net banking, UPI, wallets.
- **Cloud services:** storage and software on demand (Google Drive, AWS).
- **Social media:** Facebook, Instagram, YouTube, LinkedIn for promotion.

Intranet and Extranet

- **Intranet:** private network inside a company for staff (policies, HR portal).
- **Extranet:** intranet extended to selected outsiders - suppliers, dealers.

Tools used on the Internet

- **Browser:** software to view web pages - Chrome, Firefox, Edge.
- **URL:** address of a page, like https://www.example.com.
- **ISP:** company giving internet access - Jio, Airtel, BSNL.
- **Domain name:** easy name for an IP address (DNS converts it).
- **Web server:** computer that stores the site and serves pages.



8. Online Shopping

Online Shopping:

Buying goods or services directly from a seller over the Internet using a web browser or mobile app (also called e-tailing).

Steps in Online Shopping



- **Search:** find a product by category, name or filter.
- **Compare:** price, features, ratings and seller.
- **Cart:** items stored until purchase.
- **Payment:** card, UPI, net banking, wallet or Cash on Delivery (COD).
- **Order tracking:** SMS / email updates with a tracking id.
- **Return & refund:** as per the return policy.

Advantages of Online Shopping

- Convenient, saves time, no crowds, open 24 x 7.
- More variety and lower prices; offers and festival sales.
- Easy comparison; user reviews.
- Gifts can be sent to another city.

Disadvantages

- Cannot try or touch; size / colour mismatch.
- Risk of fraud, fake products, data leak.
- Delivery delay and shipping cost.
- Returns can be difficult.

Popular Indian Examples

Type	Examples
General marketplace	Amazon India, Flipkart, Meesho
Fashion	Myntra, Ajio, Nykaa
Food & grocery	Swiggy, Zomato, BigBasket, Blinkit
Travel & tickets	IRCTC, MakeMyTrip, RedBus
Government e-market	GeM (Government e-Marketplace)

9. E-Commerce Opportunities for Industries

Meaning:

E-commerce gives every industry new ways to sell, to cut cost, to reach customers and to serve them better.

Industry opportunities

Retail	Banking & finance	Travel & tourism	Education
online stores, marketplaces, D2C brands	net banking, UPI, e-wallets, insurance	ticket, hotel, tour booking	online courses, e-books, MOOCs

Industry-wise Opportunities

- **Retail & FMCG:** online outlets, home delivery, quick commerce, direct-to-consumer brands.
- **Banking & insurance:** internet banking, mobile apps, online policy purchase and claims.
- **Travel & hospitality:** booking flights, trains, hotels; dynamic pricing; reviews.
- **Education & training:** virtual classrooms, online degrees, e-certification.
- **Healthcare:** tele-consultation, e-pharmacy, online reports, health apps.

- **Manufacturing:** online ordering of raw materials, supplier portals, B2B trade, direct sale to customers.
- **Agriculture:** e-NAM, online mandi prices, selling produce directly, farm input buying.
- **Entertainment & media:** OTT streaming, e-books, music apps, online gaming, digital news.
- **Real estate:** property listing, virtual tours, online rent / booking.
- **Government:** e-governance, online tax, GeM, DigiLocker, e-tendering.
- **Logistics:** tracking, warehouse automation, last-mile delivery apps.
- **Small & medium business:** easy selling through marketplaces, social media shops, ONDC.

Benefits to Industry

- New markets and more revenue sources.
- Lower marketing and operating cost.
- Faster product cycles and quick feedback.
- Better supplier and dealer coordination.
- Mass customisation - products tailored to each customer.

Challenges for Industry

- Security and trust issues; need for skilled staff.
- Logistics, packaging and return management.
- Legal rules - GST, consumer protection, IT Act 2000.

10. Quick Revision - Unit I

- E-business = all business activities done electronically; e-commerce = online buying / selling only.
- Functions of EC: information, ordering, payment, delivery, support, marketing.
- Advantages: reach, 24 x 7, low cost, convenience. Disadvantages: security, trust, no physical check.
- Internet services: email, WWW, FTP, chat, VoIP, search, cloud, social media.
- Online shopping flow: search → cart → checkout → payment → delivery → feedback.
- Opportunities: retail, banking, travel, education, healthcare, manufacturing, agriculture, media, government.

Likely Exam Questions

- Define e-business. Explain its nature and significance.
- Distinguish between e-commerce and e-business.
- Explain advantages and disadvantages of e-commerce.
- Describe the internet services useful for e-business.
- Discuss e-commerce opportunities for industries.

Learning Outcomes Check

- I can explain the concept of e-business.
- I can compare e-commerce and e-business.
- I can evaluate e-commerce opportunities for industry.

I. Introduction to Electronic Markets

Electronic Market (E-Market):

An online place where buyers and sellers meet, exchange information, negotiate prices and complete transactions through the Internet, without being physically present.

Meaning and Features

- Works like a traditional market, but the market place is a website or app.
- Buyers and sellers can be individuals, firms or governments.
- Open 24 x 7 and reaches customers anywhere.
- Prices are transparent; comparison is easy.
- Fewer intermediaries → lower cost (disintermediation).
- Some new intermediaries appear (marketplaces, payment gateways) - called re-intermediation.

Functions of an Electronic Market

- **Matching:** brings buyers and sellers together.
- **Information:** product details, price, availability.
- **Transaction support:** order, payment, delivery tracking.
- **Trust & legal framework:** ratings, escrow, dispute resolution.

Electronic Market vs Traditional Market

Basis	Traditional Market	Electronic Market
Place	Physical location	Website / mobile app
Time	Fixed shop hours	24 x 7
Reach	Local	National / global
Cost	High rent & staff	Low overhead
Price comparison	Difficult	Instant
Payment	Cash mostly	Cards, UPI, wallets

★ Types of e-markets covered here: E-Shop, E-Mall, E-Grocery and Portals.

2. E-Shops

E-Shop:

A website through which a single company or retailer markets and sells its own products or services directly to customers online. (Also called online store, web shop.)

- Run by ONE business - the brand sells its own goods.
- Has its own domain name, catalogue, shopping cart and payment gateway.
- Examples: Titan.co.in, Nike.com, Apple Store online, Tanishq.

Features of an E-Shop

- Product catalogue with photos, price, description.
- Shopping cart and wish-list.
- Secure checkout and multiple payment options.
- Customer account, order history, tracking.
- Offers, coupons and email newsletters.
- Support through chat, phone, FAQ.

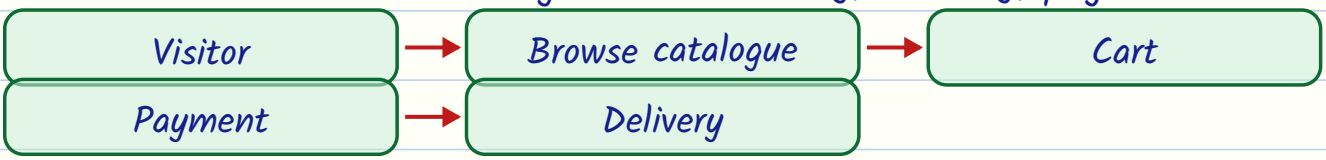
Advantages of E-Shops

- Direct contact with customers - no middlemen commission.
- Full control over brand, price and design.
- Customer data is owned by the seller.
- Low cost compared to opening a physical shop.

Disadvantages

- Seller must create traffic through marketing and SEO.
- Customers may not trust a new unknown shop.

- Needs own technical management - hosting, security, payments.



3. E-Malls

E-Mall:

An online shopping centre / marketplace that hosts many independent e-shops under one web address, offering a wide range of products from different sellers.

- Like a shopping mall, but virtual: one site → many shops.
- Mall owner provides platform, traffic, common payment system and delivery support; earns commission or listing fee.
- Examples: Amazon, Flipkart, Myntra, Snapdeal, Meesho.

Types of E-Malls

- **Referral mall:** only gives links to individual shops; each shop does its own selling.
- **Managed mall:** mall owner handles payment, order, delivery for all shops.

Features of E-Malls

- Many categories of products in one place.
- Common shopping cart and single payment.
- Search, filters, ratings and reviews.
- Seller dashboard to manage listings and orders.
- Offers like festival sales and bank discounts.

E-Shop vs E-Mall

Basis	E-Shop	E-Mall
Sellers	Only one	Many independent sellers
Products	Own brand products	Variety of brands / categories
Owner	Seller owns site	Mall operator owns platform
Traffic	Seller must attract	Mall brings ready traffic
Revenue	Product sales	Commission, ads, fees
Example	Nike.com	Amazon, Flipkart

Advantages of E-Malls

- One-stop shopping; easy comparison.
- Small sellers get instant reach and trust of mall brand.
- Shared infrastructure reduces cost.

Disadvantages

- Heavy competition among sellers; commission cuts profit.
- Seller depends on mall policies.

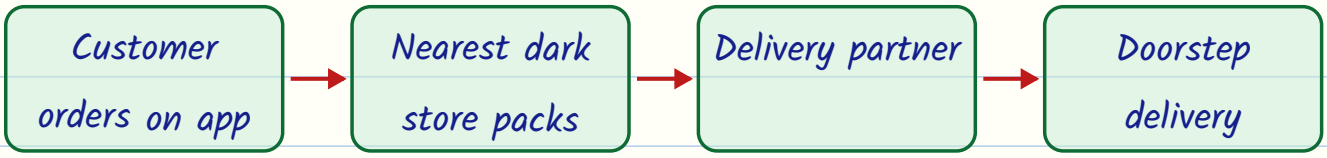
4. E-Groceries

E-Grocery:

Online buying of food and household items such as vegetables, fruits, dairy, staples and FMCG, delivered to the customer's home or collected from a store.

- Examples in India: BigBasket, Blinkit, Zepto, Swiggy Instamart, JioMart, Amazon Fresh, DMart Ready.

Working Model



Features

- Large product list with fresh and packed items.
- Slot-based, same-day or 10 - 15 minute quick delivery.
- Subscription for milk and daily needs.
- Cash on delivery or digital payment.
- Easy re-order list and offers.

Advantages

- Saves travel and queue time; helpful for senior citizens.
- Price and brand comparison; discounts.
- Reduces wastage through demand planning.

Challenges

- Fresh produce quality and freshness.
- Cold chain, packaging and last-mile delivery cost.
- Low margins and high competition.
- Returns of perishable goods.

5. Portals

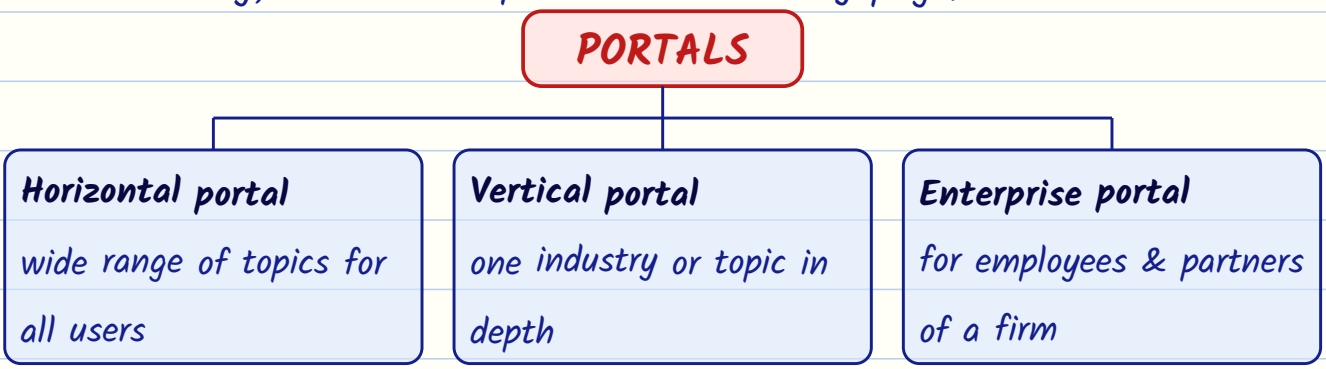
Portal:

A web site that acts as a gateway or starting point to the Internet and offers a wide range of services - search, news, email, shopping, entertainment - in one place.

- Examples: Yahoo, MSN, Rediff, AOL, Naukri, MakeMyTrip, Indiatimes.

Common Services of a Portal

- Search engine and directory.
- News, weather, sports.
- Email and chat.
- Online shopping, classifieds.
- Finance, jobs, matrimony, travel.
- Community, forums and personalisation (My page).



Horizontal Portal (Broad Portal)

Horizontal Portal:
Offers a broad range of content and services on many topics for a general audience - a "mass-market gateway".

- Examples: Yahoo, MSN, Google homepage, Rediff, AOL.
- Services: search, news, mail, shopping, entertainment, finance.
- Revenue: advertising, sponsorship, premium services.

Vertical Portal (Vortal)

Vertical Portal:
Focuses on ONE particular industry, subject or interest and provides deep, specialised information and services to a niche audience.

- Examples: Naukri (jobs), Shaadi.com (matrimony), MakeMyTrip (travel), PolicyBazaar (insurance), Agriwatch (agriculture), IndiaMart (B2B trade).

- Revenue: subscriptions, listing fees, commissions, targeted ads.

Horizontal Portal vs Vertical Portal

Basis	Horizontal Portal	Vertical Portal
Scope	Wide - many subjects	Narrow - one industry / topic
Audience	General public	Specific niche audience
Depth	Shallow info	Detailed expert info
Examples	Yahoo, MSN, Rediff	Naukri, PolicyBazaar
Advertising	General ads	Highly targeted ads
Traffic	Very high	Smaller but loyal
Competition	Search giants, large portals	Few niche competitors

Advantages of Portals

- Single entry point - saves time for users.
- Personalisation of content and services.
- Large audience attracts advertisers (revenue).
- Integration of many services (mail, news, shopping).
- Community building and customer loyalty.
- Vertical portals build expertise and trust in a niche.

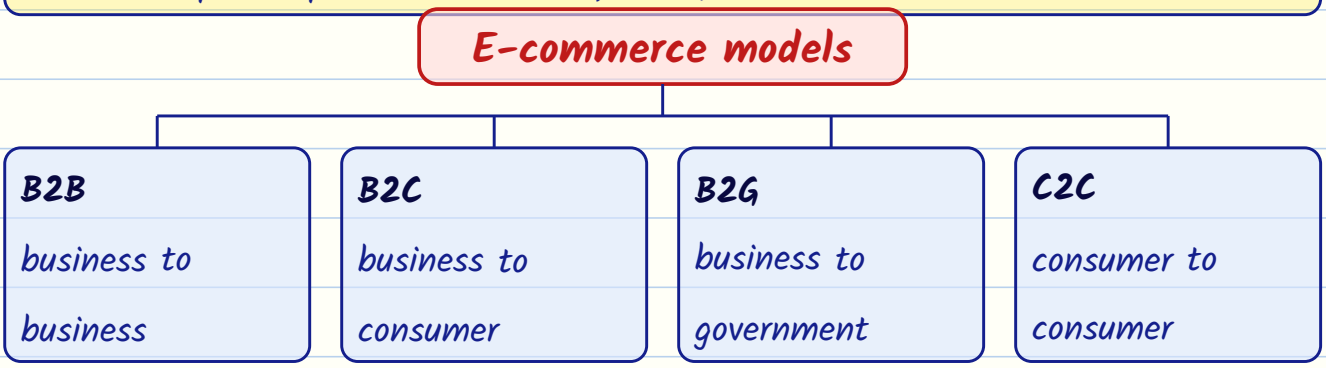
Limitations

- Information overload; slow pages.
- Dependence on ads; privacy concerns.

6. Business Models in E-Commerce

Business Model:

The method by which a company creates value, delivers it to customers and earns revenue / profit. In e-commerce it also answers who the participants are (B2B, B2C, B2G ...).



Revenue Sources in E-Business Models

- Sales of products / services.
- Commission or transaction fee.
- Advertising and sponsorship.
- Subscription and membership fees.
- Listing / registration fee.
- Data / affiliate marketing income.

7. Business to Business (B2B)

B2B:

Electronic commerce in which one business sells goods or services to another business - raw materials, components, machinery, services - mostly in bulk.

- Largest part of e-commerce by value.
- Buyers are professional; decisions on price, quality, credit terms.
- Examples: IndiaMart, Alibaba, TradeIndia, Udaan, Ariba, GeM (for govt. buyers).

Features

- Large order volume and high value.
- Long-term contracts and repeat orders.
- Negotiated, customised pricing; credit period.
- Integration with ERP, EDI and supply chain.

Types of B2B Models

- **Supplier-oriented:** one seller offers to many buyers (e.g. Dell for businesses).
- **Buyer-oriented:** one big buyer invites bids from many suppliers (e-procurement, e-tendering).
- **Intermediary-oriented:** third-party marketplace links many buyers & sellers (IndiaMart, Alibaba).



Advantages of B2B

- Lower procurement cost; wider supplier choice.
- Faster order cycle; less paperwork and errors.
- Better inventory control (just-in-time).
- Transparent pricing and tracking.

Disadvantages

- Setting up systems is costly; staff training needed.
- Trust and quality assurance with unknown suppliers.
- Security of business data.

B2B Portals in India

Portal	Focus

IndiaMART	Largest Indian B2B marketplace, SME suppliers
TradeIndia	Product listing for manufacturers & exporters
ExportersIndia	Export - import trade leads
Udaan	B2B trading for retailers (FMCG, apparel)
Ofbusiness	Raw material & credit for SMEs
GeM	Government e-Marketplace for public procurement
Moglix / Infra.Market	Industrial & construction supplies

- India also has ONDC (Open Network for Digital Commerce) as an open network for buyers and sellers.

8. Business to Consumer (B2C)

B2C:
A business sells products or services directly to individual end consumers over the Internet.

- Examples: Amazon, Flipkart, Myntra, Nykaa, Zomato, BookMyShow, Netflix.

Features

- Large number of small-value transactions.
- Mass marketing and ready-made catalogue prices.
- Quick payment by card / UPI / wallet.
- Emphasis on user experience, offers and delivery speed.

Types of B2C Models

- **Direct seller / e-shop:** manufacturer sells to consumer (Apple Store).

- **E-tailer / online retailer:** e.g. Amazon.
- **Service provider:** banking, tickets, OTT, education.
- **Content provider:** news, music, videos, e-books.
- **Community / social model:** Facebook, Instagram with ads.
- **Portal / intermediary:** MakeMyTrip, PolicyBazaar.

Advantages of B2C

- Direct reach to consumers; no middlemen.
- 24 x 7 selling; wide market; personalised offers.
- Customer feedback and data for improvement.

Disadvantages

- Intense competition and price wars.
- High delivery, returns and marketing cost.
- Need to build trust; fraud and payment risks.

B2B vs B2C

Basis	B2B	B2C
Customer	Business / organisation	Individual consumer
Order size	Bulk, high value	Small, single items
Decision	Rational, group-based	Emotional, quick
Relationship	Long-term contracts	Short-term / repeat
Price	Negotiated	Fixed listed price
Sales cycle	Long	Short
Examples	IndiaMART, Alibaba	Amazon, Myntra

9. Business to Government (B2G)

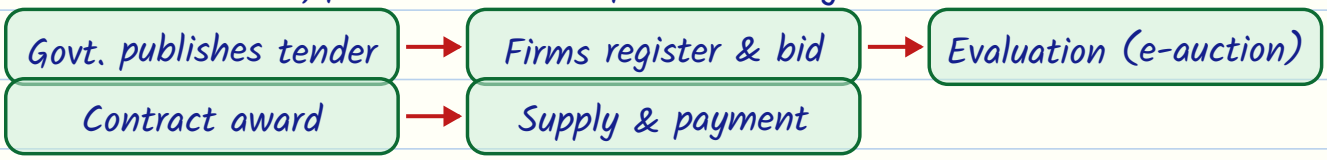
B2G:

Business-to-Government e-commerce in which companies sell goods and services to government departments and public-sector bodies through online platforms - e-tendering, e-procurement, e-filing.

- Government is the BUYER; business is the SELLER / service provider.
- Examples: GeM (gem.gov.in), CPPP e-tender portal, state e-procurement, MCA21 filing, GST portal, Income Tax e-filing, Customs ICEGATE.

Activities in B2G

- Online tender notice, bid submission and e-auction.
- Vendor registration and verification.
- Digital signature for bids; e-payment of tax and fees.
- Online licence, permit and compliance filing.



Advantages

- Transparency, less corruption and favouritism.
- Faster, cheaper procurement; open to small firms (MSME, start-ups).
- Less paperwork; audit trail is automatic.

Challenges

- Complex rules and documentation; digital-signature need.
- Security of confidential bids; technical skills of vendors.

★ G2B and G2C are the reverse: government giving services to businesses and citizens (e-governance).

10. Auctions

Online Auction:

A process of buying and selling goods through bidding on the Internet; the item goes to the highest bidder (or lowest in reverse auction) within a fixed time.

- Examples: eBay, Quikr (bidding), GeM reverse auction, MSTC e-auction, bank-property auctions, Bidgela.

Types of Auctions

- **English auction (forward):** price starts low and rises; highest bid wins.
- **Dutch auction:** price starts high and falls until a buyer accepts.
- **Sealed-bid (first-price):** bids are secret; highest bid wins.
- **Vickrey (second-price):** highest bidder wins but pays the second-highest price.
- **Reverse auction:** many sellers compete; lowest price wins (used in B2B / B2G procurement).
- **Double auction:** buyers and sellers both bid - stock exchanges.



Advantages of Online Auctions

- Real market price discovery; wide bidder base.
- Fast sale of surplus or rare items.
- Lower cost for buyers in reverse auctions.
- Open and transparent.

Disadvantages

- Fraud, shill (fake) bidding, non-payment.
- Winner's curse - overpaying.
- Needs trust, escrow and rules.

Revenue of Auction Sites

- Listing fee, final value commission, advertising, payment service charges.

II. Quick Revision - Unit II

- E-shop = single seller; E-mall = many shops under one roof; E-grocery = online food and daily needs.
- Portal = gateway site. Horizontal = broad topics; Vertical = niche industry.
- B2B = firm to firm; B2C = firm to consumer; B2G = firm to government.
- Auction types: English, Dutch, sealed-bid, Vickrey, reverse, double.
- B2B portals in India: IndiaMART, TradeIndia, Udaan, ExportersIndia, GeM, Ofbusiness.

Likely Exam Questions

- Distinguish between e-shop and e-mall.
- Compare horizontal and vertical portals; advantages of portals.
- Explain B2B, B2C and B2G business models with examples.
- Explain types of online auctions.
- Write about B2B portals in India.

I. Introduction to Electronic Payment Systems (EPS)

Electronic Payment System:

A system that allows money to be paid or transferred electronically – without cash or cheques – between buyer and seller, using computers, networks, banks and payment instruments.

Meaning

- Payment is made by electronic message, card, app or digital token.
- Needed in e-commerce because buyer and seller are not face to face.
- Involves customer, merchant, issuing bank, acquiring bank, payment gateway and a clearing network.

Features of a Good EPS

- **Security:** protects money and data from fraud.
- **Authentication:** verifies identity of payer (PIN, OTP, biometrics).
- **Integrity & confidentiality:** amount and details cannot be altered or read by others.
- **Convenience & speed:** quick, easy, anytime.
- **Low transaction cost:** suitable for small payments (micro-payments).
- **Acceptability:** widely accepted by merchants and banks.
- **Anonymity / privacy:** in some systems (cash-like).
- **Reliability & interoperability:** works across banks and apps.

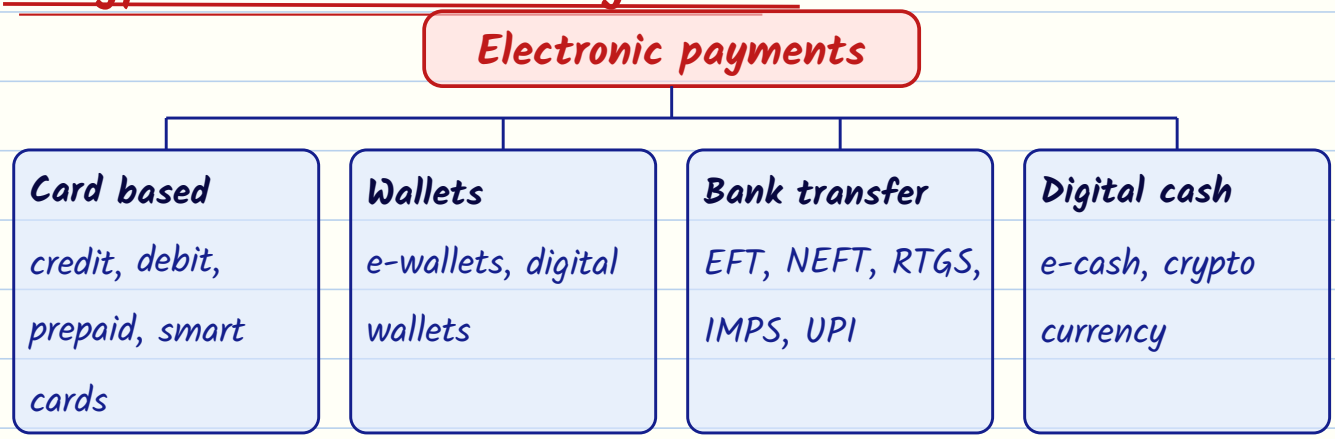
Advantages of EPS

- Faster payment and settlement; reduced cash handling.
- Transparent record of every transaction.
- Supports online shopping and global trade.
- Less risk of theft, fake notes and counting errors.
- Supports Digital India and financial inclusion.

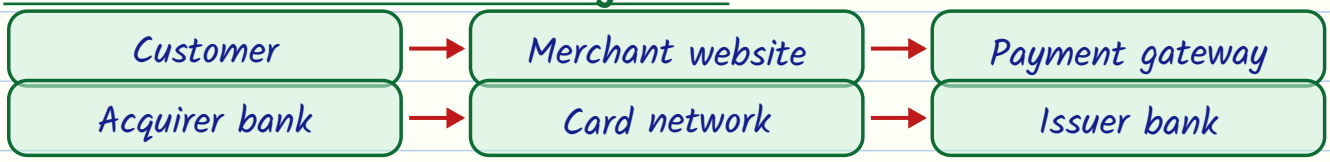
Disadvantages

- Cyber fraud, phishing, card cloning.
- Needs internet and a smartphone; technical failure.
- Transaction charges in some modes.
- Privacy - spending is tracked.

2. Types of Electronic Payments



Parties in an Online Card Payment



- **Payment gateway:** secure link that carries card data (Razorpay, PayU, CCAvenue, Paytm).
- **Acquirer bank:** merchant's bank.
- **Issuer bank:** customer's bank that gave the card.
- **Card network:** Visa, Mastercard, RuPay - route and settle the payment.

3. Credit Cards

Credit Card:

A plastic / virtual card issued by a bank that lets the holder buy now and pay later, within a sanctioned credit limit; the bank lends the money.

How It Works

- Bank sets a credit limit and billing cycle (about 30 days).
- Customer swipes / enters card details; merchant is paid by the bank.
- Customer receives a monthly statement and pays the bill by the due date.
- If only the minimum amount is paid or paid late, heavy interest (about 36 - 42 % p.a.) is charged.

Features

- Card number (16 digits), expiry date, CVV, name; chip and PIN or OTP.
- Interest-free credit period (up to about 50 days).
- Reward points, cashback, EMI options, travel / lounge benefits.
- International acceptance.

Advantages

- Buy first, pay later; emergency credit.
- Rewards, discounts and EMI facility.
- Builds credit history (CIBIL score).
- Purchase protection and dispute facility.

Disadvantages

- Overspending and debt trap; high interest and late fees.
- Annual fee and hidden charges.
- Fraud risk if details leak.

4. Debit Cards

Debit Card:

A card linked to the holder's savings / current account; the purchase amount is deducted directly from the account balance at the time of payment.

- Issued by banks: Visa, Mastercard, RuPay (e.g. SBI, HDFC, PNB cards).
- Used at ATM, POS terminals and online with PIN / OTP.

Features

- Instant debit; no borrowing.
- Daily withdrawal and spending limits.
- Works with 2-factor authentication.
- Contactless (tap and pay, NFC) in many cards.

Advantages

- No debt or interest; spending within balance.
- Easy cash withdrawal and cashless shopping.
- Usually no joining fee; simple to get.

Disadvantages

- No credit period; money leaves the account at once.
- Fraud directly affects bank balance.
- Fewer rewards and protection than credit cards.
- Payment fails if balance is low.

Credit Card vs Debit Card

Basis	Credit Card	Debit Card
Money used	Bank's money (loan)	Own money in account
Payment	Later - bill each month	Immediate deduction
Limit	Credit limit by bank	Account balance / daily limit

Interest	Charged if unpaid	None
Eligibility	Income and credit score check	Any account holder
Debt risk	High	Nil
Rewards	More rewards and benefits	Few offers
Fraud loss	Bank money first, dispute possible	Own money is lost first
Credit score	Improves with timely payment	No effect

★ Prepaid cards / gift cards: money is loaded first and used later - no credit, no bank account needed.

5. Smart Cards

Smart Card:
A plastic card with an embedded microchip (microprocessor + memory) that can store and process data securely for identity, payment and access.

Types

- **Contact smart card:** chip touches the reader (EMV chip cards, SIM).
- **Contactless card:** works by radio waves (RFID / NFC) - metro cards, tap-and-pay.
- **Hybrid / dual-interface:** both contact and contactless.

Features

- Stores money, personal data, keys and certificates in the chip.
- Strong security - PIN, encryption, tamper-proof chip.
- Offline transaction possible.
- Reusable and rechargeable (stored-value cards).

Uses

- Banking: EMV debit / credit cards.
- Transport: metro, bus, toll (FASTag uses RFID).
- Identity: Aadhaar, e-passport, employee ID.
- Telecom: SIM; health card; campus card; loyalty cards.

Advantages

- Safer than magnetic stripe; hard to clone.
- Quick tap payments; reduces cash.
- Many applications on one card.

Disadvantages

- Card reader needed; cost of issue.
- Lost card = loss of stored balance.
- Compatibility problems between systems.

6. E-Wallets and Digital Wallets

E-Wallet (Digital Wallet):
A software / app-based prepaid account that stores money or payment details electronically so that the user can pay, transfer and receive money using a phone or computer.

- Examples: Paytm, PhonePe wallet, Amazon Pay, MobiKwik, Freecharge, Google Pay, Apple Pay, Samsung Pay.

Types

- **Closed wallet:** only for the issuer's own products (Amazon Pay balance).
- **Semi-closed wallet:** many merchants, but no cash withdrawal (Paytm, Mobikwik).
- **Open wallet:** issued by banks; allows cash withdrawal and bank

transfer.

Features and Usage

- Load money by card, net banking or UPI.
- Pay via QR code, mobile number, NFC.
- Bill payment, recharge, ticket booking.
- Cashback, coupons, wallet-to-wallet transfer.
- KYC (Know Your Customer) needed for higher limits - regulated by RBI.

Advantages

- Fast, no need to enter card details each time.
- Offers and cashback; easy splitting of bills.
- Works for small value payments.

Disadvantages

- Limit on balance and transfers.
- Needs smartphone and internet; risk if phone is lost.
- Money may get stuck if the wallet company fails.

7. Smart Cards vs Digital Wallets

Basis	Smart Card	Digital Wallet
Form	Physical chip card	Software / mobile app
Storage	Chip in the card	Cloud / phone
Internet	Not always required	Usually required
Security	PIN + chip encryption	PIN, OTP, biometrics
Use	Offline and online	Mostly online / QR
Loss risk	Physical card loss	Phone loss
Examples	EMV card, metro card	Paytm, PhonePe, Google Pay

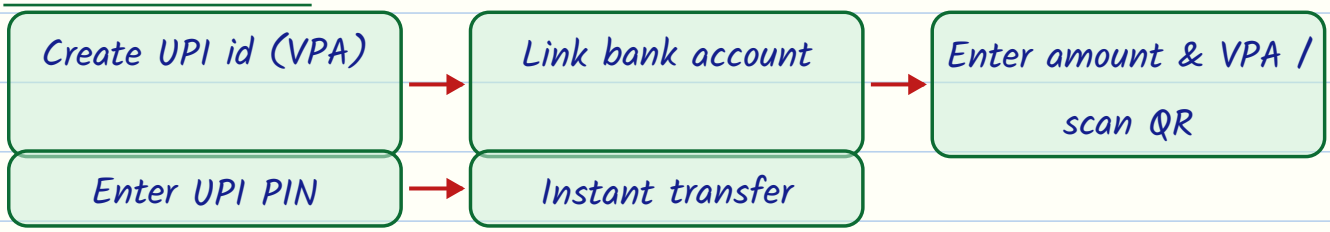
8. UPI - Unified Payments Interface

UPI:

A real-time payment system developed by NPCI (National Payments Corporation of India) that instantly transfers money between bank accounts through a mobile app using a Virtual Payment Address (VPA).

- Launched in 2016 by NPCI under RBI guidance; works 24 x 7 x 365.
- Apps: PhonePe, Google Pay, Paytm, BHIM, Amazon Pay and bank apps.

How UPI Works



Features

- Single app for many bank accounts; one mobile number = identity.
- Pay by VPA (name@bank), QR code, phone number.
- Send and request money (collect request).
- Two-factor security: device binding + UPI PIN.
- Free for most peer-to-peer payments.
- UPI Lite, UPI 123PAY (feature phones), international UPI.

Advantages

- Instant, simple, 24 x 7; no need to share account number or card.
- Interoperable between all banks; low cost.
- Supports merchants, bill pay, mandates (auto-pay).

Disadvantages

- Needs internet; server downtime sometimes.
- Fraud through fake collect requests and QR scams.
- Per-transaction limits (usually Rs. 1 lakh).

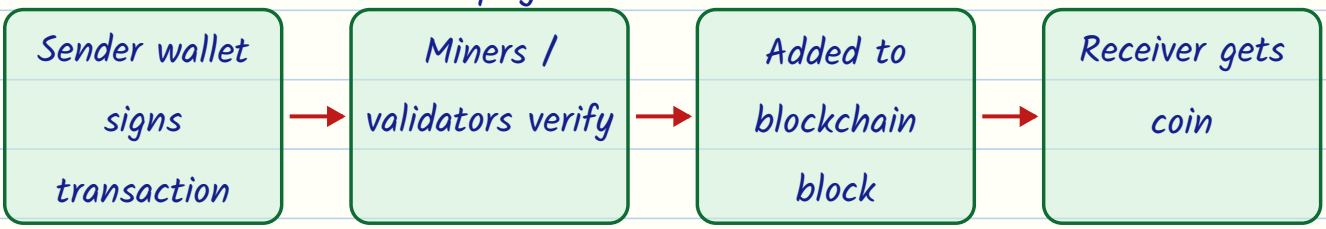
9. Crypto Currencies

Cryptocurrency:
A digital / virtual currency secured by cryptography, created and recorded on a decentralised blockchain, with no central bank or government control.

- Examples: Bitcoin (2009), Ethereum, Ripple (XRP), Litecoin, Tether (stable coin).

Features

- **Decentralised:** no single authority; peer-to-peer network.
- **Blockchain:** public ledger of all transactions, in linked blocks.
- **Cryptographic security:** public / private keys, hashing.
- **Limited supply:** e.g. Bitcoin max 21 million.
- **Pseudonymous & global:** transfers anywhere in minutes.
- **Irreversible:** confirmed payment cannot be reversed.



Advantages

- Low-cost cross-border payments; no middlemen.
- Strong security and transparency.
- Open to the unbanked.

Disadvantages

- Very volatile price; speculation risk.
- Used for illegal activity; scams and exchange hacks.
- Lost private key = lost coins.
- Regulation is uncertain; in India it is not legal tender - gains taxed at 30 % + 1 % TDS; RBI is developing the digital rupee (e-Rupee, a CBDC).

10. Electronic Fund Transfer (EFT)

EFT:

The electronic transfer of money from one bank account to another - within one bank or between banks - using computer networks, without paper cheques or cash.

Role of EFT in Business Transactions

- Salary and wage payment (payroll), pension, dividends.
- Supplier / vendor payments and customer receipts.
- Tax payment (GST, income tax), utility and loan EMIs.
- E-commerce settlement between buyer, seller and payment gateway.
- Reduces cash handling, float time, paperwork and errors.

Modes of EFT in India

Mode	Key points
NEFT	Batch based, 24 x 7; no minimum limit; small to medium amounts
RTGS	Real-time gross settlement; minimum Rs. 2 lakh; large value
IMPS	Instant 24 x 7 via mobile / net banking
UPI	Instant, via VPA / QR; PIN based

ECS / NACH	Bulk recurring credit / debit - salary, EMI, SIP
ATM / POS	Card based cash withdrawal and purchase

Advantages

- Fast and safe; lower cost; easy record keeping and audit.
- Reduces cheque bouncing and delays.

Disadvantages

- Wrong account number can cause wrong credit; reversal is difficult.
- Needs bank account and internet; cyber risks.

Smart Card vs EFT (Learning Outcome)

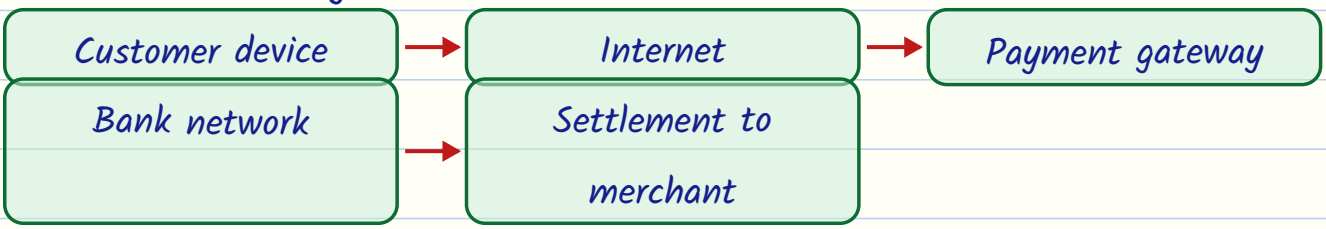
Basis	Smart Card	EFT
Nature	A payment instrument / device	A method of moving funds between accounts
Medium	Chip card with stored data	Bank network (NEFT, RTGS, IMPS)
Use	Point-of-sale, transport, ID	Salary, vendor, large business payments
Value	Small to medium	Small to very large
Money source	Stored value or linked account	Directly from bank account
Offline use	Possible	Needs network

II. Infrastructure Requirements for E-Payments

- **Hardware:** servers, POS machines, ATMs, smartphones, card readers, NFC / QR scanners.
- **Software:** payment gateway, banking software (CBS), merchant

apps, shopping-cart software.

- **Networks:** reliable internet, VPN / leased lines, SWIFT, NPCI network (UPI, IMPS, RuPay).
- **Security tools:** SSL / TLS, encryption, firewalls, tokenisation, OTP / biometrics, digital certificates.
- **Banking set-up:** merchant account, acquiring bank, issuer banks, clearing house.
- **Standards:** EMV chip standard, PCI-DSS (card data security), ISO 8583 messages.
- **Support:** call centre, dispute / chargeback system, backup and disaster recovery.



12. Regulatory Aspects of E-Payments

Key Authorities and Laws in India

- **RBI (Reserve Bank of India):** regulates banks, wallets (PPI guidelines), card rules, two-factor authentication, tokenisation of cards.
- **NPCI:** operates UPI, IMPS, RuPay, NACH, FASTag.
- **Payment and Settlement Systems Act, 2007:** legal base for payment systems; RBI authorises operators.
- **Information Technology Act, 2000:** recognises electronic records and digital signatures; cyber-crime offences.
- **Consumer Protection Act / E-Commerce Rules 2020:** protects

online buyers.

- **PMLA & KYC norms:** anti-money laundering checks.
- **Data protection:** Digital Personal Data Protection Act, 2023; RBI data storage rules.
- **PCI-DSS:** international security standard for card data.

Consumer Protection Measures

- Report unauthorised transaction quickly: zero liability if reported within 3 working days (as per RBI rules).
- Two-factor authentication, OTP, card tokenisation.
- Grievance redressal through bank, RBI Ombudsman and NPCI.

★ Tip: Never share OTP, PIN, CVV or UPI PIN with anyone.

13. Quick Revision - Unit III

- EPS = payment without cash. Types: cards, wallets, UPI, EFT, crypto, smart cards.
- Credit card = pay later (bank loan); debit card = pay now from own account.
- Smart card = chip card; EFT = account-to-account bank transfer (NEFT, RTGS, IMPS).
- E-wallet = prepaid app (closed, semi-closed, open). UPI = instant bank-to-bank via NPCI.
- Crypto = blockchain based decentralised digital money; risky and not legal tender in India.
- Infrastructure: servers, gateway, network, security, banks. Regulators: RBI, NPCI, IT Act, PSS Act.

Likely Exam Questions

- Explain types of electronic payment systems.
- Distinguish between credit card and debit card.
- Compare smart cards and EFT.
- Explain UPI and digital wallets with features.
- Discuss infrastructure and regulatory aspects of e-payments.

I. Introduction to E-Security

E-Security:

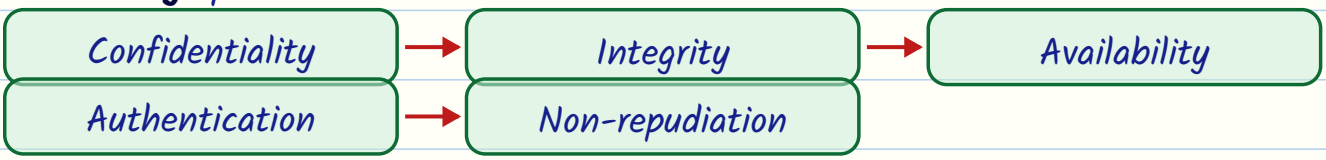
The set of policies, technologies and practices used to protect e-commerce systems, data, networks and transactions from unauthorised access, misuse, theft and damage.

Why Security is Needed in E-Commerce

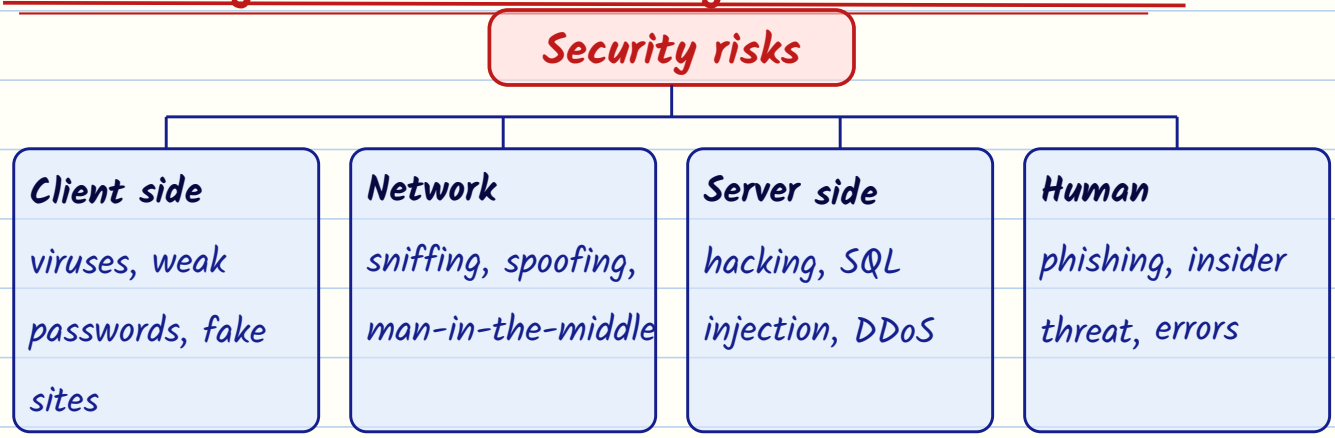
- Money and sensitive data (card numbers, passwords, addresses) travel over open networks.
- Buyer and seller cannot see each other - trust is needed.
- Hackers, fraudsters and malware attack 24 x 7.
- Business loses money, reputation and customers after a breach.
- Laws (IT Act 2000, DPDP Act 2023) require protection of data.

Security Requirements (CIA + more)

- **Confidentiality:** only authorised people can read the data (encryption).
- **Integrity:** data is not changed during transfer (hashing, digital signature).
- **Availability:** system is up when needed (backup, anti-DDoS).
- **Authentication:** proving who the user is (password, OTP, biometrics).
- **Authorisation:** what the user is allowed to do (access control).
- **Non-repudiation:** sender cannot deny sending (digital signature).
- **Privacy:** personal information is not misused.



2. Security Risks and Challenges in E-Commerce



Types of Risks

- **Data theft / breach:** card and customer data stolen from servers.
- **Payment fraud:** stolen cards, chargeback fraud, fake orders.
- **Unauthorised access:** weak or stolen passwords.
- **Denial of Service (DoS / DDoS):** site flooded with traffic and goes down.
- **Malicious software:** viruses, worms, ransomware.
- **Spoofing & sniffing:** fake identity; capturing network data.
- **Website defacement and fake web sites.**
- **Insider threats:** employees misuse access.

Challenges

- Rapidly changing attack methods; need for skilled staff.
- Balance between security and ease of use.
- Cost of security tools for small firms.
- Cross-border crime and weak law enforcement.
- Mobile apps, cloud and IoT widen the attack surface.
- Customer awareness is low.

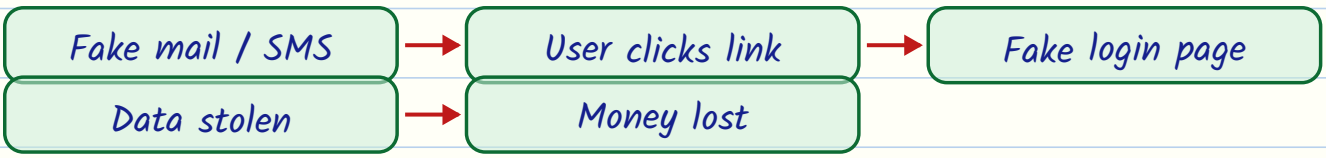
3. Cyber Threats

Cyber Threat:
Any possible malicious act that tries to damage, steal or disrupt data and digital systems through the Internet or networks.

3.1 Phishing

Phishing:
A fraud in which attackers send fake emails, SMS or web links that look like they are from a trusted organisation (bank, courier, e-shop) to trick the user into giving passwords, OTP or card details.

- Types: email phishing, spear phishing (targeted), whaling (top executives), smishing (SMS), vishing (voice call), pharming (fake site by redirecting DNS).
- Signs: urgent message, spelling mistakes, strange link or sender, asks for OTP / PIN.



- **Prevention:** do not click unknown links; check URL and https; never share OTP; use spam filter and 2-factor authentication.

3.2 Hacking

Hacking:
Gaining unauthorised access to a computer, network or website by exploiting weaknesses, to steal, change or destroy data.

- **Black-hat hacker:** criminal intent. **White-hat (ethical) hacker:** tests security legally. **Grey-hat:** in between.
- Methods: password guessing / brute force, SQL injection, cross-site scripting (XSS), exploiting software bugs, DDoS, social engineering.

- **Prevention:** strong passwords, patching, firewalls, intrusion detection, penetration testing.

3.3 Identity Theft

Identity Theft:

Stealing someone's personal information - name, Aadhaar, PAN, bank or card details - and using it to commit fraud such as opening accounts, taking loans or shopping in the victim's name.

- Ways: phishing, data breach, skimming card devices, shoulder surfing, dumpster diving, fake KYC calls.
- Effects: financial loss, bad credit score, legal problems, loss of reputation.
- **Prevention:** protect documents; use strong unique passwords; check bank statements; enable alerts; avoid public Wi-Fi for payments; freeze / lock cards when lost.

3.4 Malware

Malware:

Malicious software designed to harm a computer, steal data or give control to an attacker.

Type	Description
Virus	Attaches to files / programs and spreads when they are run
Worm	Self-copies across networks without user action
Trojan horse	Looks useful but hides harmful code; opens backdoor
Ransomware	Locks / encrypts files and demands money (ransom)
Spyware	Secretly records activity, passwords, keystrokes
Adware	Shows unwanted ads; may track users

Botnet	Network of infected devices controlled remotely (used for DDoS)
Keylogger	Records keys typed to steal passwords

- **Prevention:** antivirus and updates, avoid pirated software and unknown attachments, regular backups, least-privilege accounts.

Other Threats (brief)

- **DoS / DDoS:** overloading a site with requests.
- **Man-in-the-middle (MITM):** attacker secretly sits between two parties and reads / changes messages.
- **Spoofing:** pretending to be another user, site or IP address.
- **Sniffing:** capturing data packets on a network.
- **SQL injection:** inserting harmful database commands through input forms.
- **Social engineering:** tricking people into revealing secrets.

4. Cryptography - Basis of Security

Cryptography:

The science of converting readable data (plaintext) into unreadable form (ciphertext) and back, so that only the intended person can read it.



Symmetric Key (Secret Key) Encryption

- Same key for encryption and decryption.
- Fast; good for large data. Problem: how to share the key safely.

- Algorithms: DES, 3DES, AES, RC4.

Asymmetric Key (Public Key) Encryption

- Two keys - a PUBLIC key (shared with all) and a PRIVATE key (kept secret).
- Data encrypted with one key can be decrypted only by the other.
- Slower but solves key sharing; used for digital signatures and key exchange.
- Algorithms: RSA, ECC, Diffie-Hellman.

Symmetric vs Asymmetric

Basis	Symmetric	Asymmetric
Keys	One shared secret key	Public + private key pair
Speed	Fast	Slow
Key distribution	Difficult	Easy (public key is open)
Use	Bulk data encryption	Signatures, key exchange
Example	AES, DES	RSA, ECC

Hash Function

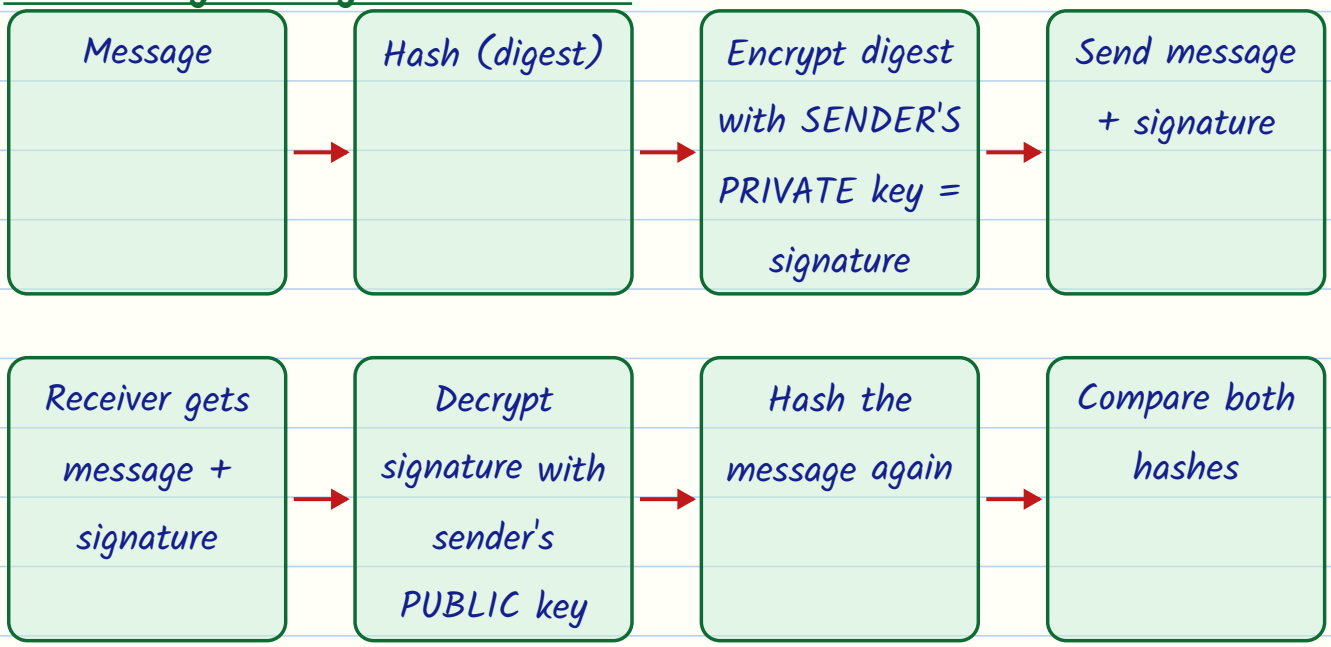
- Converts any data into a fixed-length "fingerprint" (message digest); one-way; tiny change gives completely different hash.
- Used to check integrity and in signatures. Examples: SHA-256, MD5 (old).

5. Digital Signatures

Digital Signature:

An electronic signature created with the sender's private key on the hash of a message; it proves WHO sent the message (authentication), that it was NOT changed (integrity) and that the sender cannot deny it (non-repudiation).

How a Digital Signature Works



- If both hashes match → message is genuine and unchanged.
- If not → message was altered or sender is fake.

Features / Advantages

- Authentication, integrity, non-repudiation.
- Legally valid under the IT Act 2000.
- Saves paper and time; used in e-tendering, e-filing, GST, MCA, banking.

Limitations

- Private key theft is dangerous; certificate cost and expiry.
- Needs trusted Certifying Authority and software support.

★ Digital signature is NOT the same as a scanned signature image or a typed name.

6. Digital Certificates

Digital Certificate:

An electronic document issued by a trusted Certifying Authority (CA) that binds a person's / website's identity to their public key, like an online ID card.

Contents of a Certificate (X.509)

- Name of owner and organisation; owner's public key.
- Serial number; validity period (start and end date).
- Name and digital signature of the issuing CA.
- Algorithm used; certificate class.

Certifying Authorities

- In India the CCA (Controller of Certifying Authorities) licenses CAs - eMudhra, Sify, nCode, (n)Code Solutions, NIC CA, IDRBT, Capricorn.
- International: DigiCert, GlobalSign, Sectigo, Let's Encrypt.

Classes of Certificate (India)

- **Class 1:** for individuals, checks email / name only.
- **Class 2:** for individuals and companies; identity verified (used for filing e-forms).
- **Class 3:** highest level; physical verification; for e-tendering, e-auction, e-commerce sites.

Uses

- Digitally sign documents and tenders; secure email.
- SSL / TLS certificate for HTTPS websites.
- Code signing; VPN and device authentication.



Public Key Infrastructure (PKI)

- Framework of people, policies, hardware and software to create, manage, distribute and revoke digital certificates.
- Components: Certifying Authority (CA), Registration Authority (RA), certificate repository / directory, CRL (Certificate Revocation List), key management.

Digital Signature vs Digital Certificate

Basis	Digital Signature	Digital Certificate
Purpose	Proves message origin and integrity	Proves identity of key owner
Created by	Sender using private key	Certifying Authority
Contains	Encrypted hash of message	Public key + identity + CA signature
Validity	For that message	For a fixed period (1 - 3 years)

7. Security Protocols over Public Networks

A security protocol is a set of rules that protects data while it travels across the open Internet.

7.1 HTTP and HTTPS

- **HTTP (HyperText Transfer Protocol):** rules for sending web pages between browser and server; sends data as PLAIN TEXT, so not secure; port 80.

- **HTTPS (HTTP Secure):** HTTP running on top of SSL / TLS; data is encrypted; shows padlock in browser; port 443.

Basis	HTTP	HTTPS
Security	None - plain text	Encrypted with SSL / TLS
Port	80	443
URL	http://	https://
Certificate	Not needed	Needed (SSL certificate)
Use	Simple info sites	Banking, shopping, login pages

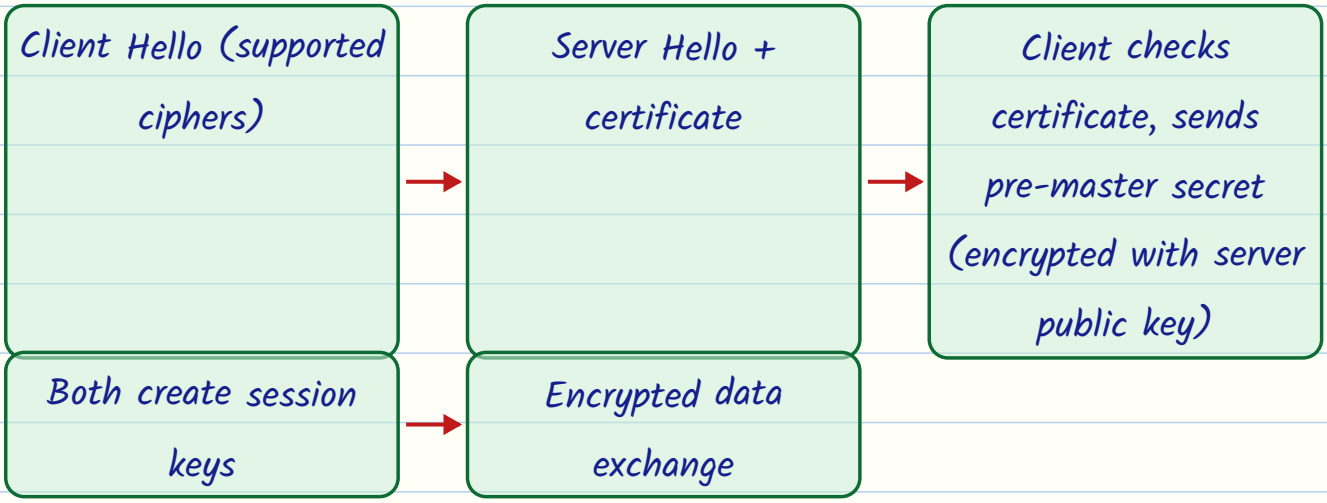
7.2 SSL - Secure Sockets Layer

SSL:

A protocol developed by Netscape (1990s) that creates an encrypted link between a web server and a browser so that data passed between them stays private and unaltered.

- Works between TCP and application layer (HTTP, FTP, SMTP).
- Provides: encryption (confidentiality), message integrity (MAC), server authentication (certificate), optional client authentication.
- SSL 2.0 / 3.0 are now obsolete due to weaknesses (POODLE attack).

SSL Handshake (How a Secure Session Starts)



- Asymmetric cryptography is used in the handshake; fast symmetric keys are used for the actual data.

7.3 TLS - Transport Layer Security

TLS:

The modern, improved successor of SSL, standardised by IETF; it secures communication over public networks and is used in HTTPS, email and VPN.

- Versions: TLS 1.0, 1.1 (old), TLS 1.2 (widely used), TLS 1.3 (fastest and safest).
- Stronger ciphers and hashing; faster handshake in TLS 1.3.
- Two layers: TLS Record protocol (encrypts data) and TLS Handshake protocol (negotiates keys).

SSL vs TLS

Basis	SSL	TLS
Developer	Netscape	IETF
Status	Deprecated	Current standard
Versions	2.0, 3.0	1.0, 1.1, 1.2, 1.3
Security	Weaker, known flaws	Stronger algorithms
Handshake	Slower	Faster (1.3)

7.4 SET - Secure Electronic Transaction

- Protocol by Visa and Mastercard for card payments: uses digital certificates and dual signatures so merchant cannot see card number and bank cannot see order details. (Rarely used now; replaced by SSL / TLS + 3-D Secure.)

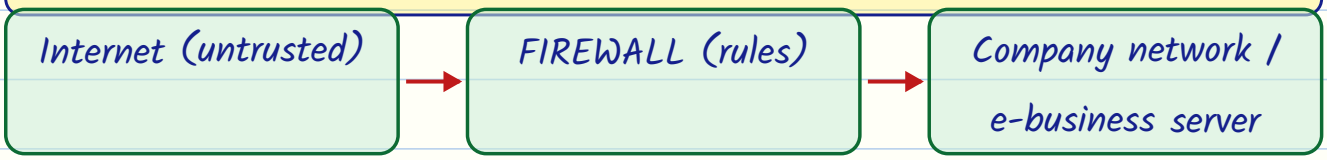
Other Security Methods

- **3-D Secure / OTP:** extra password for card payments (Verified by Visa).
- **Two-factor authentication (2FA):** password + OTP / biometric.
- **Tokenisation:** card number replaced with a random token.
- **VPN, IPSec, SSH:** secure remote links and logins.

8. Firewalls

Firewall:

A hardware or software security system placed between a trusted internal network and an untrusted network (Internet) that monitors and controls incoming and outgoing traffic according to security rules.



Functions

- Allows or blocks traffic by IP address, port, protocol or application.
- Hides internal network (NAT); logs and reports suspicious activity.
- Prevents unauthorised access, malware entry and some DoS attacks.
- Creates a DMZ - a buffer zone holding public web servers.

Types of Firewalls

- **Packet-filtering firewall:** checks each packet's header (IP, port); fast, simple, but cannot see content.
- **Stateful inspection firewall:** tracks connection state, allows only valid replies.
- **Application-level (proxy) firewall:** works at application layer;

inspects content; most secure but slower.

- **Circuit-level gateway:** checks session set-up (TCP handshake).
- **Next-generation firewall (NGFW):** deep packet inspection, intrusion prevention, app control.
- **Hardware vs software firewall:** appliance for network vs program on a PC.

Advantages

- First line of defence; central control of traffic.
- Stops many attacks; supports policy and logging.

Limitations

- Cannot stop insider attack or infected USB / email attachments.
- Cannot protect if user gives away the password; needs regular rule updates.
- Not a replacement for antivirus, IDS and encryption.

Other Protective Tools

- Antivirus / anti-malware, IDS / IPS, regular patching, backups, strong password policy, employee awareness, security audit.

9. Security Best Practices for E-Business

- Use HTTPS with a valid SSL / TLS certificate on every page.
- Use payment gateways that follow PCI-DSS; never store CVV.
- Enable two-factor authentication for admin and customers.
- Keep servers, plugins and apps updated; take regular backups.
- Limit access by role (least privilege); log and monitor.
- Train staff and customers about phishing.
- Prepare an incident-response and disaster-recovery plan.

Legal Framework

- **IT Act 2000 (amended 2008):** legal recognition of e-records and digital signatures; punishes hacking (Sec. 66), identity theft (66C), cheating by impersonation (66D), cyber terrorism (66F).
- **Controller of Certifying Authorities (CCA)** and CERT-In (incident response).
- **DPDP Act 2023:** protection of personal data.
- Report cyber crime: cybercrime.gov.in or helpline 1930.

10. Quick Revision - Unit IV

- Security needs: confidentiality, integrity, availability, authentication, non-repudiation.
- Threats: phishing, hacking, identity theft, malware (virus, worm, trojan, ransomware, spyware).
- Encryption: symmetric (one key) vs asymmetric (public + private).
- Digital signature = hash encrypted with sender's private key; certificate = CA-signed binding of identity and public key.
- HTTP plain text; HTTPS = HTTP + SSL / TLS; TLS is the modern replacement for SSL.
- Firewall filters traffic between trusted and untrusted networks.

Likely Exam Questions

- Explain security risks and challenges in e-commerce.
- Describe phishing, hacking, identity theft and malware.
- What is a digital signature? Explain how it works and its advantages.
- Explain digital certificates and PKI.
- Discuss HTTP, SSL and TLS; compare SSL and TLS.
- What is a firewall? Explain types and functions.

I. Introduction to E-Marketing

E-Marketing (Online Marketing):

Promoting and selling products, services and brands using the Internet and digital channels - websites, search engines, social media, email, mobile apps - and measuring the results.

Meaning

- Also called internet marketing, digital marketing, web marketing.
- Uses the 4 Ps (product, price, place, promotion) in a digital form.
- Two-way communication: customers can reply, share, review.
- Results are measurable in real time (clicks, views, sales).

Online Marketing Mix (E-Marketing 4 Ps -> 4 Cs)

Traditional P	Online view
Product	Customisable, digital products, bundles
Price	Dynamic pricing, discounts, price comparison
Place	Website, app, marketplace - available everywhere
Promotion	SEO, ads, social media, email, influencers

Advantages of Online Marketing

- **Global reach:** from local to worldwide.
- **Low cost:** cheaper than TV or print; small budgets possible.
- **Targeting:** by age, location, interest and behaviour.
- **Measurable:** analytics show clicks, conversions and ROI.
- **Interactive & personal:** engage and build relations with customers.
- **24 x 7 availability** and instant updates.
- **Quick feedback** and A/B testing of offers.
- **Level playing field:** small firms compete with large ones.

Limitations

- Heavy competition and ad fatigue; ad blockers.
- Privacy and spam complaints; click fraud.
- Needs technical skill and constant updating.
- Not reachable to people without internet.

2. Internet Advertisement

Internet Advertising:

Paid promotional messages delivered to users through websites, search engines, apps, email and social media in order to attract attention and create sales.

Features

- Interactive - users can click, watch, buy at once.
- Targeted and personalised; cost per click / view.
- Real-time measurement; flexible budget and timing.

Advantages

- Lower cost and wider reach than traditional media.
- Immediate results; campaigns can be paused or changed anytime.

Disadvantages

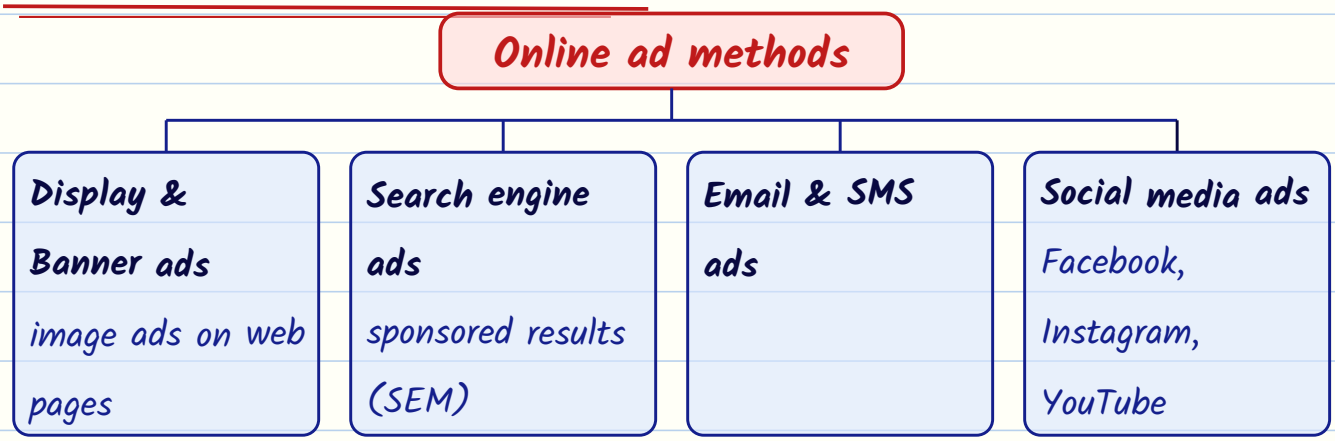
- Ad blockers; users ignore banners (banner blindness).
- Privacy concerns about tracking; click fraud.

Pricing Models of Internet Ads

Model	Meaning
CPM (Cost per Mille)	Pay for every 1000 impressions (views)
CPC / PPC (Pay per Click)	Pay each time a user clicks the ad
CPA / CPL (per Action / Lead)	Pay only when a sale, signup or lead happens

CPV (per View)	Pay for video views (YouTube)
Flat fee / sponsorship	Fixed price for a time period or position

3. Advertisement Methods



3.1 Banner / Display Ads

- Graphic or animated ads on web pages; clicking leads to the advertiser site.
- Sizes: leaderboard, skyscraper, rectangle. Retargeting shows ads again to past visitors.

3.2 Search Engine Marketing (SEM)

- **SEO (Search Engine Optimisation):** improve site content, keywords, speed and links to rank higher in FREE (organic) results.
- **PPC (Google Ads):** paid sponsored results on top of search page; pay per click.

3.3 Email Advertising

- Promotional mails and newsletters to subscribers; high return on investment.
- Needs permission (opt-in) to avoid spam.

3.4 Social Media Ads

- Paid posts and videos on Facebook, Instagram, YouTube, LinkedIn, X; very precise targeting.

3.5 Pop-up, Interstitial and Video Ads

- Pop-up: new small window; interstitial: full screen between pages; video: pre-roll / mid-roll on YouTube.

3.6 Other Methods

- **Affiliate marketing:** partners promote your product and earn commission per sale.
- **Influencer marketing:** popular personalities endorse the brand.
- **Native / sponsored content:** ad that looks like normal article or post.
- **Classified ads:** OLX, Quikr, Justdial listings.
- **Mobile advertising:** in-app ads, SMS, push notifications.
- **Viral marketing:** message spreads on its own through sharing.
- **Content & blogs, webinars, podcasts, QR codes.**

4. Conducting Online Market Research

Online Market Research:

Collecting, analysing and interpreting information about customers, competitors and the market using the Internet, to support marketing decisions.

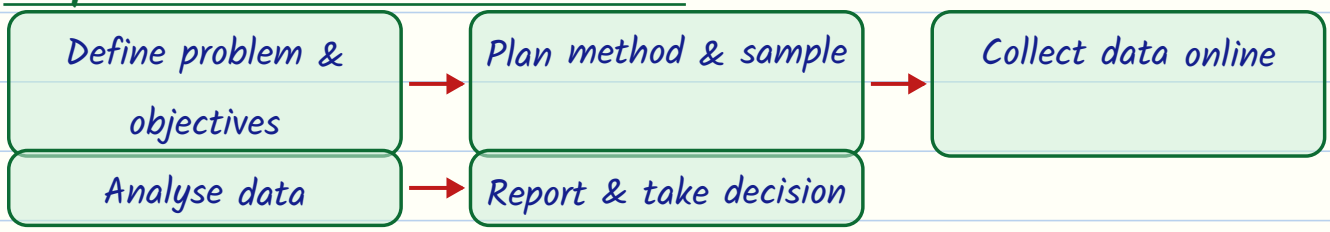
Types of Data

- **Primary data:** gathered first-hand for the purpose.
- **Secondary data:** already published (reports, websites, government data).

Methods of Online Research

- **Online surveys / questionnaires:** Google Forms, SurveyMonkey, email polls.
- **Focus groups and interviews:** by video call or chat rooms.
- **Web analytics:** page views, bounce rate, time spent, sources of traffic (Google Analytics).
- **Social listening / monitoring:** track brand mentions, trends, sentiment.
- **Customer reviews and feedback forms.**
- **Experiments:** A/B testing of pages and ads.
- **Secondary research:** search engines, online databases, competitor sites, e-journals, government portals (data.gov.in).
- **Cookies and clickstream data** - tracking user behaviour.

Steps in Online Market Research



Advantages

- Fast, cheaper and wider reach than field research.
- Easy data entry and automatic analysis; real-time results.
- Respondents answer at convenient time; fewer interviewer errors.

Limitations

- Sample bias - only internet users take part; low response rates.
- Fake or careless responses; privacy and consent issues.
- Cannot observe body language.

5. E-Marketing Planning

An e-marketing plan sets goals for online activities, defines the target audience, selects channels and budget, and sets measures of success.



- **Objectives:** brand awareness, leads, sales, retention - make them SMART (specific, measurable, achievable, relevant, time-bound).
- **Target market:** segmentation by demographics, interests, behaviour; build customer personas.
- **Positioning:** how the brand is different from competitors.
- **Tactics:** SEO, ads, content, email, social, mobile.
- **Budget, schedule and responsibility.**
- **Control:** KPIs - traffic, conversion rate, cost per acquisition, ROI, customer lifetime value.

5.1 Online Branding

Online Branding:

Building a recognisable identity, reputation and trust for a business on the Internet - name, logo, domain, website design, tone of content and customer experience.

- Choose a short memorable domain name and logo; consistent colours and message everywhere.
- Quality content, fast and mobile-friendly website, good user experience.

- Customer reviews, testimonials and quick support build trust.
- Brand presence on social media, marketplaces and search results.
- Monitor and manage online reputation (ORM).

5.2 Social Media Marketing (SMM)

SMM:

Using social platforms - Facebook, Instagram, YouTube, LinkedIn, X, WhatsApp - to create awareness, interact with customers and drive traffic and sales.

- Activities: regular posts, reels / videos, stories, contests, live sessions, groups and communities, paid ads, influencer tie-ups, customer service replies.
- Benefits: low cost, viral reach, direct engagement, precise targeting, instant feedback.
- Risks: negative comments, fake news, time and constant effort, platform algorithm changes.
- Measures: likes, shares, comments, reach, followers, click-through rate (CTR).

5.3 Email Marketing

Email Marketing:

Sending commercial messages - newsletters, offers, order updates, reminders - to a list of people who have agreed to receive them.

- Types: welcome mail, promotional, newsletter, abandoned-cart reminder, transactional, re-engagement.
- **Best practices:** permission-based list, catchy subject line, personalisation (name, interest), clear call-to-action (CTA), mobile-friendly design, test and send at the right time, easy unsubscribe.

- Measures: open rate, click-through rate, bounce rate, conversion rate, unsubscribes.
- Advantages: very cheap, high ROI, direct, easy to automate.
- Problems: spam filters, low open rates, privacy law (consent).

Comparison of the Three Channels

Basis	Online Branding	Social Media	Email
Aim	Identity and trust	Engagement and reach	Direct sales and retention
Cost	Medium	Low - medium	Very low
Reach	Wide	Very wide	Existing list
Key tool	Website, logo, ORM	Facebook, Instagram, YouTube	Mailing list, newsletters

6. E-Business Strategies

E-Business Strategy:

A long-term plan that explains how a firm will use digital technology and the Internet to create value, gain competitive advantage and reach its business goals.

6.1 Digital Advertising

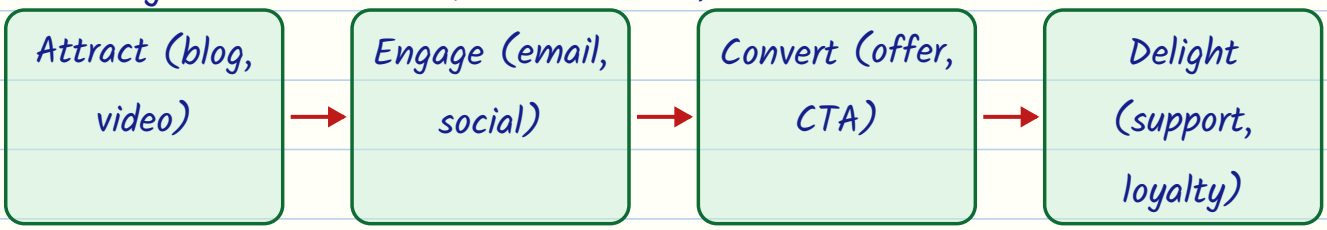
- Paid media: search ads, display, video, social ads, mobile and programmatic (automated bidding in real time).
- Needs clear target, creative message, landing page and tracking of conversions.
- Retargeting and remarketing increase sales from interested visitors.

6.2 Content Marketing

Content Marketing:

Creating and sharing valuable, relevant and consistent content - blogs, videos, podcasts, infographics, e-books, webinars - to attract and keep a clearly defined audience and finally drive profitable action.

- Content builds trust, helps SEO, educates customers and nurtures leads.
- Steps: know audience → plan content calendar → create → distribute (social, email, search) → measure and improve.
- Types: educational (how-to), entertaining, inspirational, user-generated (UGC), case studies, FAQs.



6.3 Analytics

Web / Digital Analytics:

Collecting, measuring and analysing data about website visitors and campaigns to understand behaviour and improve results.

- Tools: Google Analytics, Search Console, social insights, heat maps, CRM reports.
- **Metrics:** visitors / sessions, page views, bounce rate, average time on site, traffic source, click-through rate, conversion rate, cart abandonment, cost per acquisition, ROI, customer lifetime value.
- Uses: find best channel, improve website, personalise offers, decide budget, predict demand (big data and AI).

Other E-Business Strategies

- Cost leadership through automation; differentiation by service and customisation.
- Mass personalisation and recommendation engines.
- Strategic alliances, affiliate and partner programmes.
- Omni-channel presence: website + app + store + marketplace.
- Disintermediation and re-intermediation; first-mover advantage; network effects.

7. E-Customer Relationship Management (eCRM)

eCRM:

Use of the Internet and electronic tools to attract, understand, serve and retain customers by managing all interactions through web, email, chat, social media and mobile apps.

Objectives

- Acquire new customers, retain existing ones, raise customer lifetime value.
- Give personalised, quick and consistent service on every channel.

Components of eCRM

- **Operational:** automation of sales, marketing and service (help desk, chatbot, order tracking).
- **Analytical:** data warehouse, data mining to find buying patterns and segments.
- **Collaborative:** sharing customer information between departments and partners through email, portals, social media.



Features / Tools

- Customer database and profile; self-service portal and FAQs.
- Live chat, chatbots, email response management, call-centre integration.
- Loyalty programmes, personalised recommendations, targeted email.
- Feedback and complaint handling; social CRM.
- Software: Salesforce, Zoho CRM, HubSpot, Microsoft Dynamics.

Advantages

- Better customer satisfaction and loyalty; repeat sales.
- Lower service cost through self-service and automation.
- Better targeting and 360-degree view of customer.
- Quick problem solving; improved brand image.

Challenges

- High cost of set-up and integration; data quality.
- Privacy and data protection duties.
- Staff training and change in culture.
- Too much automation may lose the human touch.

CRM vs eCRM

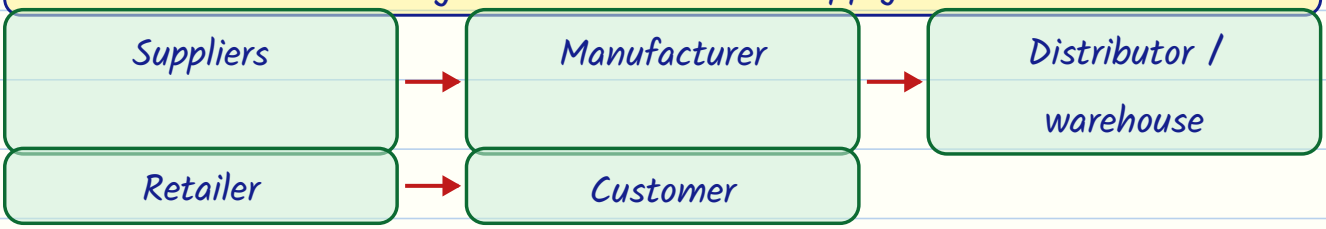
Basis	Traditional CRM	eCRM
Channel	Phone, in person, mail	Web, email, chat, social, app
Time	Business hours	24 x 7
Data	Limited and scattered	Integrated and real-time

Cost	Higher	Lower per contact
Interaction	One-way / slow	Interactive and instant

8. E-Supply Chain Management (e-SCM)

Supply Chain:

The network of suppliers, manufacturers, warehouses, distributors, retailers and customers that move a product from raw material to final consumer. e-SCM:: Use of the Internet and information technology to plan, coordinate and control the flow of materials, information and money across the whole supply chain.



- Three flows: **product flow** (forward goods and returns), **information flow** (orders, stock, tracking) and **finance flow** (payments, credit).

Components / Activities

- E-procurement: online buying, tenders and supplier portals.
- E-inventory and warehouse management (barcode, RFID).
- Demand forecasting and production planning.
- E-logistics: transport tracking (GPS), route planning, last-mile delivery.
- Order management and e-payment; returns management.

Technologies Used

- ERP, EDI (electronic data interchange), extranet, cloud platforms.
- RFID, barcode, GPS / IoT sensors, blockchain for traceability.

- Analytics, AI forecasting, automation and robots.

Advantages of e-SCM

- Lower inventory, purchase and logistics cost.
- Faster delivery and order cycle; fewer errors and stock-outs.
- Real-time visibility; better supplier and partner coordination.
- Reduced bullwhip effect (demand distortion up the chain).
- Higher customer satisfaction.

Challenges

- High IT investment; integration of partners' systems.
- Security and trust in sharing data.
- Disruptions (pandemics, wars, port delays) and dependency on technology.

eCRM vs e-SCM

Basis	eCRM	e-SCM
Focus	Customer side (front end)	Supplier and logistics side (back end)
Aim	Satisfy and retain customers	Efficient flow of goods and information
Parties	Customers	Suppliers, producers, distributors
Examples	Zoho CRM, Salesforce	SAP, Oracle SCM, EDI, GPS tracking

9. Quick Revision - Unit V

- E-marketing = promoting products through digital channels; measurable, low cost, targeted.
- Ad methods: banner, search (SEO / PPC), email, social, video,

affiliate, influencer. Pricing: CPM, CPC, CPA.

- Online market research: surveys, analytics, social listening, secondary data.
- E-marketing plan: SWOT, SMART objectives, target, mix, budget, KPIs.
- Branding builds trust; SMM engages; email is cheap and direct.
- Strategies: digital ads, content marketing, analytics. eCRM manages customers; e-SCM manages supply flows.

Likely Exam Questions

- Define online marketing and explain its advantages.
- Explain internet advertisement methods.
- How is online market research conducted?
- Explain online branding, social media marketing and email marketing.
- Describe digital advertising, content marketing and analytics strategies.
- Explain eCRM and e-SCM with features, advantages and challenges.